Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2026

Course: B.Tech
Code: 101402

Subject: Building Planning and Computer-Aided Civil Engineering Drawing

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 *Choose the correct option / answer the following (Any seven question only):* [2 x 7 = 14]

(a) In an inclusive building design, the user-centric approach primarily emphasizes
(i) Minimizing construction cost regardless of user needs
(ii) Providing identical circulation arrangements in every building
(iii) Prioritizing appearance over accessibility
(iv) Designing spaces according to the needs of diverse users

(b) Which drawing provides a vertical view of a building as if it were cut by an imaginary vertical plane?
(i) Floor plan
(ii) Site plan
(iii) Section
(iv) Terrace plan

(c) A building project requires a drawing showing the arrangement of the building within its plot, including access and surrounding features. Which drawing is most appropriate?
(i) Foundation plan
(ii) Structural section only
(iii) Door schedule
(iv) Site plan

(d) Which fire-protection arrangement is specifically intended to distribute water automatically over a fire-affected area when activated?
(i) EPBX system
(ii) Sprinkler system
(iii) Lightning conductor
(iv) Automatic water-level controller

(e) In a high-rise building, which system is intended to provide a vertical water-supply arrangement for firefighting?
(i) Ventilation duct
(ii) Domestic drainage stack
(iii) Rainwater downpipe
(iv) Wet or dry riser system

(f) Which AutoCAD command is used to draw a straight-line segment between two specified points?
(i) Circle
(ii) Offset
(iii) Line
(Iv) Trim

(g) While preparing a floor plan, you want to create a rounded corner between two intersecting wall lines with a specified radius. Which command should you use?
(i) Fillet
(ii) Explode
(iii) Chamfer
(Iv) Stretch

(h) You have drawn a rectangular room and want to create an identical room at another location without altering the original. Which command is appropriate?
(i) Break
(ii) Erase
(Iii) Copy
(Iv) Scale

(i) You need to create a parallel line exactly 230 mm away from an existing wall line in a building plan. Which command is most suitable?
(i) Rotate
(ii) Move
(Iii) Extend
(Iv) Offset

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(j) Which practice is most appropriate for sustainable water management in a residential building?

(i) Using water-efficient fixtures and collecting rainwater where feasible

(ii) Mixing stormwater with sewage without treatment

(iii) Increasing potable water consumption to improve sanitation

(iv) Discharging all wastewater directly onto the site

Q.2 (a) Explain the basic principles of residential building planning and identify the major factors governing the selection of a suitable building site. [7]

(b) Define a green building and explain its major benefits, important features and commonly used green building rating systems. [7]

Q.3 (a) Explain the fundamental principles of inclusive building design and describe the provisions required to make public buildings accessible to differently abled persons. [7]

(b) Explain the essential accessibility provisions in public sanitation facilities and discuss suitable emergency evacuation arrangements for differently abled occupants. [7]

Q.4 (a) Differentiate between load-bearing and RCC-framed structures with respect to load transfer, structural components, suitability and applications. Support your answer with neat sketches. [7]

(b) A college building is proposed on an urban site. Identify and explain the major planning requirements concerning classroom layout, circulation, sanitation, accessibility and fire safety. [7]

Q.5 (a) Develop a conceptual scheme for water supply, wastewater collection and stormwater drainage in a multistorey residential building. Explain the functions of the major components. [7]

(b) Prepare a suitable solid-waste management strategy for a multistorey residential building incorporating wet and dry waste segregation, vermicomposting and waste collection chutes. Justify the proposed arrangement. [7]

Q.6 (a) A residential complex experiences water overflow from overhead tanks and requires improved energy management and internal communication. Analyse how automatic water-level controllers, solar panels and EPBX/intercom systems can address these problems. [7]

(b) A high-rise building requires an effective fire-safety and evacuation strategy. Analyse the role of fire-safety corridors, escape routes and associated evacuation provisions, with reference to applicable building safety requirements. [7]

Q.7 (a) A multistorey building is proposed to accommodate residential and office occupants. Analyse the factors governing elevator selection, capacity, number, location, safety features and accessibility. [7]

(b) A building experiences inadequate indoor air circulation during summer. Analyse the factors affecting natural and mechanical ventilation and recommend suitable measures to improve indoor environmental conditions. [7]

Q.8 (a) A plot measuring 30 m × 40 m is proposed for residential construction. Explain how you would prepare a site plan, identifying essential features and selecting a suitable drawing scale. [7]

(b) Using AutoCAD, explain how absolute and relative coordinate systems can be used to draw a rectangular room measuring 4 m × 3 m. Write the sequence of commands and coordinate inputs required for both methods. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Green building rating systems

(b) wet and dry riser systems used in fire protection

(c) EPBX and intercom systems in buildings

(d) National Building Code in building planning

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